Stage 1: Market Data Ingestion & Exchange Synchronization
- **Official Sources**: Ingests NSE `EQUITY_L.csv` (2,639 listed stocks) & BSE corporate directory.
- **Price Feeds & Macro**: Fetches daily OHLCV price histories from Yahoo Finance & RBI repo rate bulletins.
- **Storage**: `exchange_universe_sync.py` reconciles ISINs into PostgreSQL: `securities`, `market_prices`.

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Stage 2: Investor Input & Suitability Profiling (Web UI)
- **Web Views**: Investor interacts with `OnboardingWizard.tsx` & `PortfolioGenerator.tsx`.
- **Parameters**: Inputs capital (e.g. ₹2,50,000), investment horizon, risk questionnaire answers, and sector exclusions.
- **FastAPI Routers**: `api/v1/risk_assessments.py` & `goals.py` save records to `risk_assessments`, `financial_goals`.

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Stage 3: Multi-Agent Stock Research (LangGraph 13-Node Flow)
- **Screening Filter**: `CandidateUniverseSelector` screens full catalog into 2,605 active liquid securities across 4 liquidity tiers.
- **Research Agents (Gemini 3.8)**: `macro_analyst`, `market_analyst`, `fundamental_analyst`, `news_analyst`.
- **Strategy & Risk Audit**: `analyst_node` (Lead Strategist) proposes weights; `critic_node` audits 25%

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Stage 4: Quantitative Math Optimization & Integer Lot Sizing
- **Quadratic Solver**: `optimizer.py` (SciPy SLSQP) optimizes continuous weights to maximize Sharpe ratio.
- **Whole Shares**: `allocation_rounding.py` rounds continuous percentages into discrete executable whole shares.
- **Cash & Taxes**: Preserves leftover cash reserve; `transaction_costs.py` calculates statutory Indian STT, GST, and stamp duty.

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Stage 5: Verification, Stock-Why Explanations & Database Commit
- **Final Audit**: `critic_final_validation_node` verifies whole shares satisfy all diversification constraints.
- **Explainability**: `moderator_explanation_node` & `stock_why_service.py` generate retail narrative & financial scorecard.
- **ACID Commit**: `persistence_node` commits versioned portfolio atomically to PostgreSQL: `portfolios`,

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Stage 6: Client Presentation & Continuous Market-Hours Monitoring
- **Dashboard**: Next.js renders `HoldingsView.tsx` ('My Portfolio'), `StockWhyModal.tsx`, and `StressScenariosView.tsx`.
- **Live Monitor**: `monitoring_scheduler.py` tracks live prices every 5 minutes during exchange hours (09:15-15:30 IST).
- **Drift Alert**: If holdings drift >= 5%, triggers `NotificationsPanel.tsx` and prepares `RebalancePreviewModal.tsx`.